Project 21 · CSAI

Flight Delay Prediction via Gradient Boosting

Phase 1: Machine Learning



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Can we predict a delay before the flight leaves?



Target

Arrival 15 or more minutes late, the US standard definition of a delayed flight



Decision time

Before departure (the morning of the flight day): only the schedule and past days may be used



Why it matters

Airlines can plan spare crews and aircraft, and warn passengers early

24.5%

of flights were delayed

2013 New York City departures

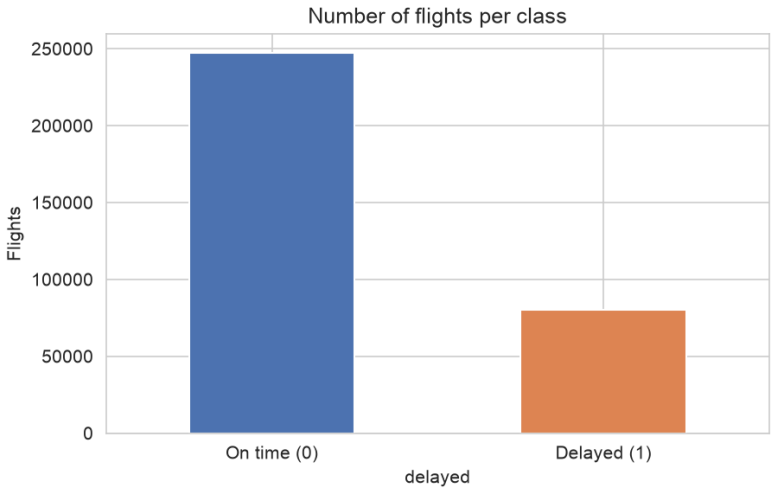
About 1 in 4, so the classes are imbalanced: accuracy is misleading, PR AUC is used instead

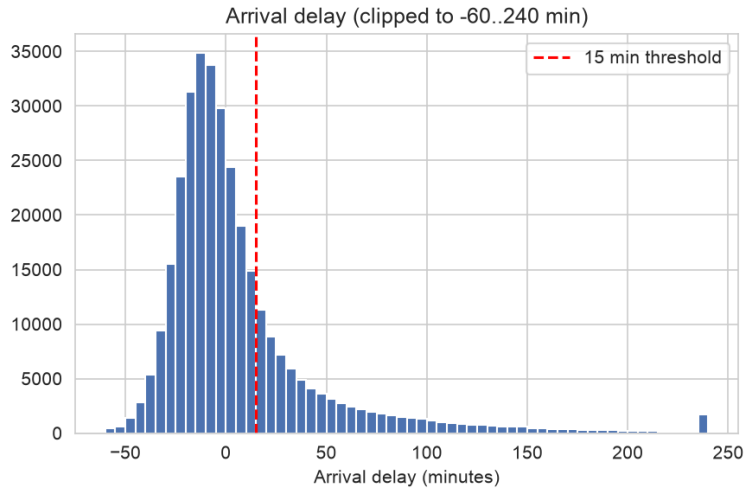
Dataset: every NYC departure in 2013

336,776
flights in the raw data

3
airports: JFK, LGA, EWR

327,346
flights kept after cleaning

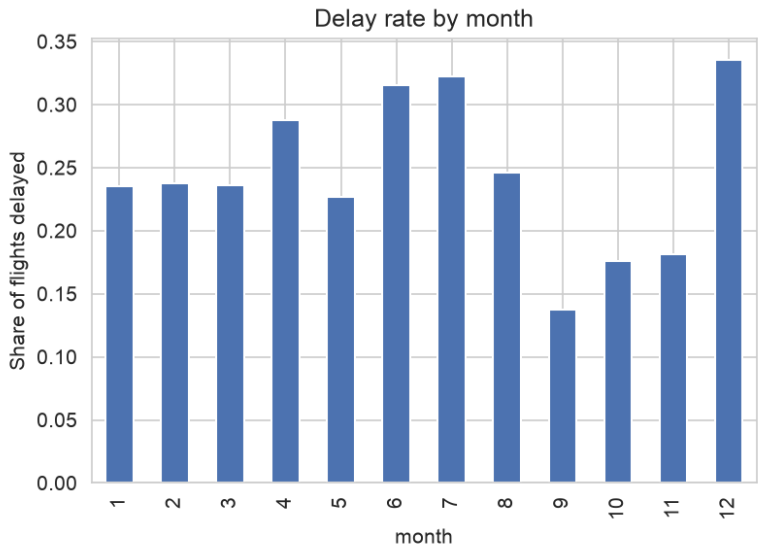


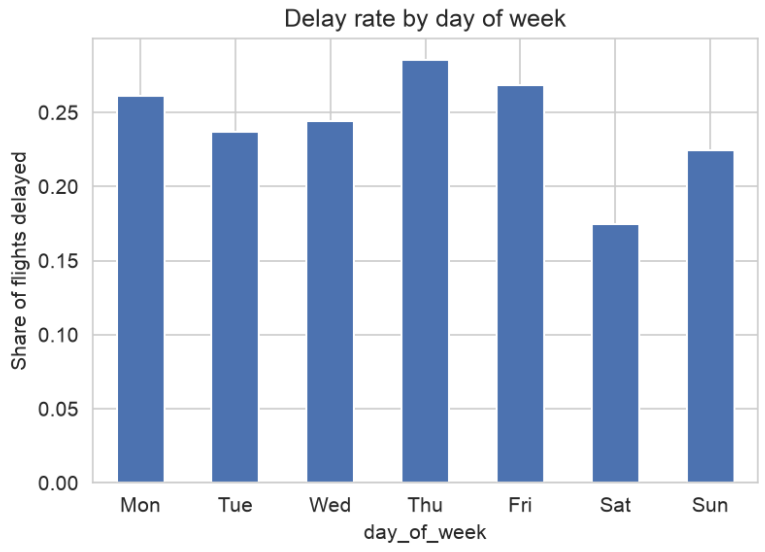


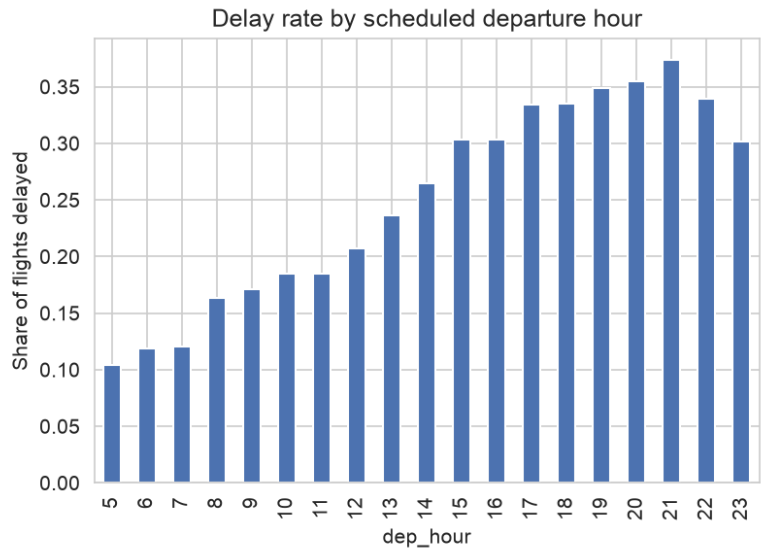
Data quality

- 9,430 flights have no arrival delay (cancelled or diverted), so they are removed
- 0 duplicate rows
- Delays have a long right tail: a few flights are hours late

Delays build up through the day







Hour of day

About 10% of 5 am flights are late, rising to about 37% at 9 pm

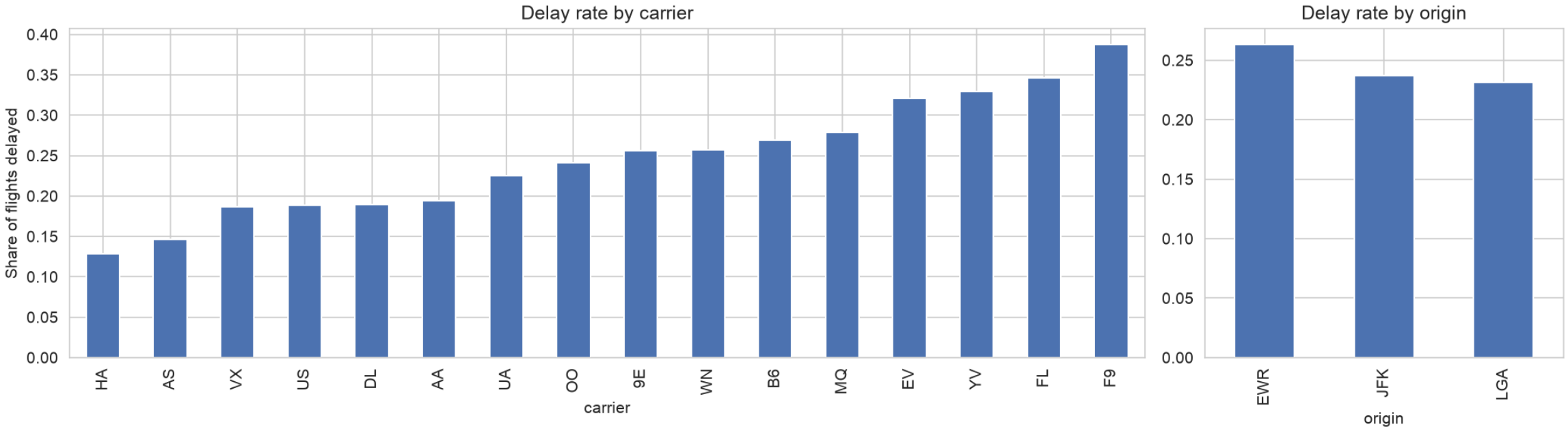
Season

Dec, Jul and Jun are worst (over 30%); Sep to Nov are calmest (under 20%)

Day of week

Saturday is the best day (about 18% delayed)

Carrier and airport matter too



Carriers

From 13% delayed (HA) to 39% (F9)

Airports

EWR is the most delayed of the three (26%)

What it means

Carrier, airport and hour are useful features

No information from after departure

Decision time: the morning of the flight day. Every feature must be known by then.

Removed (only known after departure)

- Actual departure time and delay
- Actual arrival time and delay (used only for the target)
- Air time
- Tail number and flight number (IDs)

Kept (known in advance)

- Date, scheduled departure and arrival time
- Carrier, origin, destination, distance
- The published schedule of all flights
- Outcomes of past days only

Congestion counts use the full schedule, including flights that were later cancelled. Counting only completed flights would leak the weather: bad days have many cancellations.

Four groups of pre-departure features

Calendar and schedule

Weekend flag and distance. Hour and day of week use cyclic encoding: sin(2πt/P) and cos(2πt/P), so hour 23 sits next to hour 0

Airport congestion

Flights scheduled to leave the same airport in the same hour and on the same day

Recent delay history

Share of delayed flights for the carrier, origin, destination and route over the previous 7 days, today excluded

Categories and route

Carrier, origin and destination one-hot encoded; route = origin and destination pair (223 routes)

Month is left out on purpose: the test months (Nov to Dec) never appear in training.

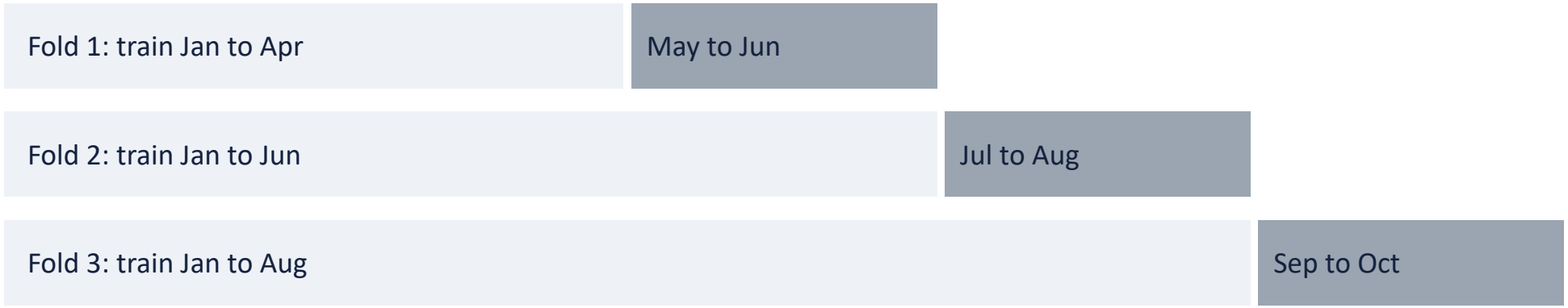
Split by date, not at random

Train
Jan to Aug
217,727 flights · 26.4% delayed

Validation
Sep to Oct
55,628 · 15.7%

Test
Nov to Dec
53,991 · 25.8%

XGBoost tuning: expanding-window time-series folds



The test months are never used for tuning. They are used once, for the final results.

A linear baseline and boosted trees

Logistic Regression

- P(delay) = σ(w·x + b)
- L2-regularised log loss
- Inputs standardised
- Balanced class weights
- Simple and interpretable baseline

XGBoost

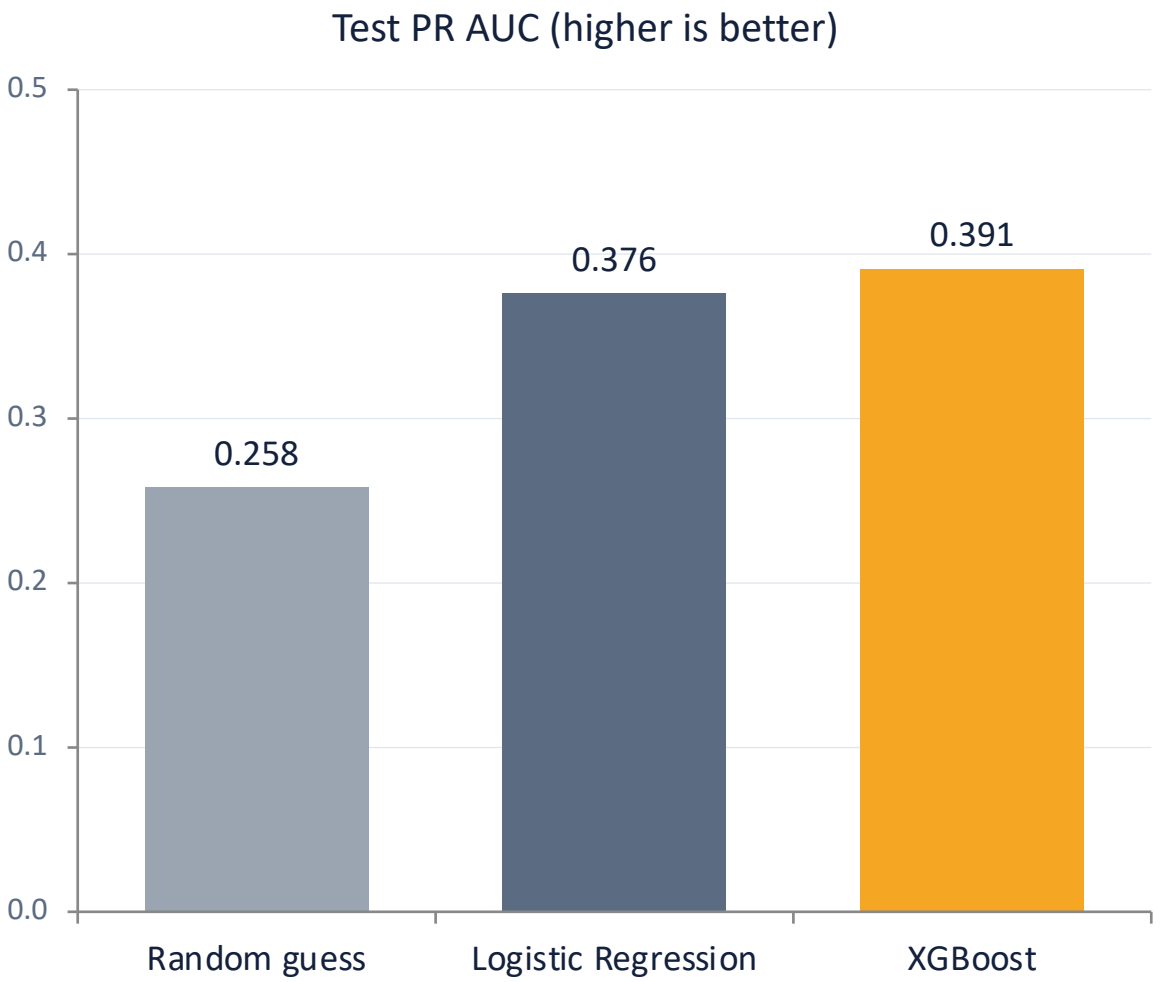
- Sum of small trees, each fixing the errors of the ones before
- Learns non-linear effects and interactions
- Learning rate 0.05, subsample 0.8, min child weight 50
- scale_pos_weight = 2.79 for imbalance

Tuning results (mean PR AUC of 3 folds)

Depth	Trees	PR AUC
3	**100**	**0.377**
4	100	0.376
6	100	0.376
3	300	0.370
4	300	0.369
6	300	0.367

Best: depth 3, 100 trees. Bigger models did worse on future months.

XGBoost is the best model on the test set

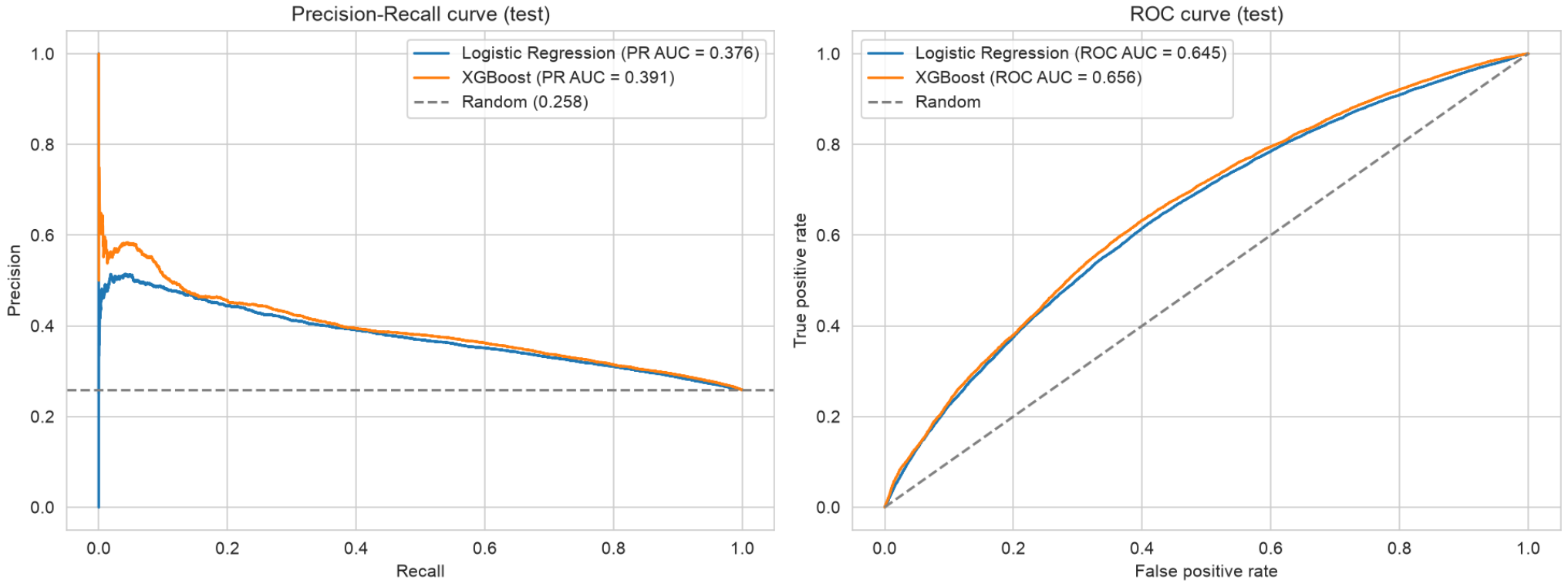


Model	PR AUC	ROC AUC	Brier
Random	0.258	0.500	-
Log. Reg.	0.376	0.645	0.228
XGBoost	**0.391**	**0.656**	**0.219**

11.1%

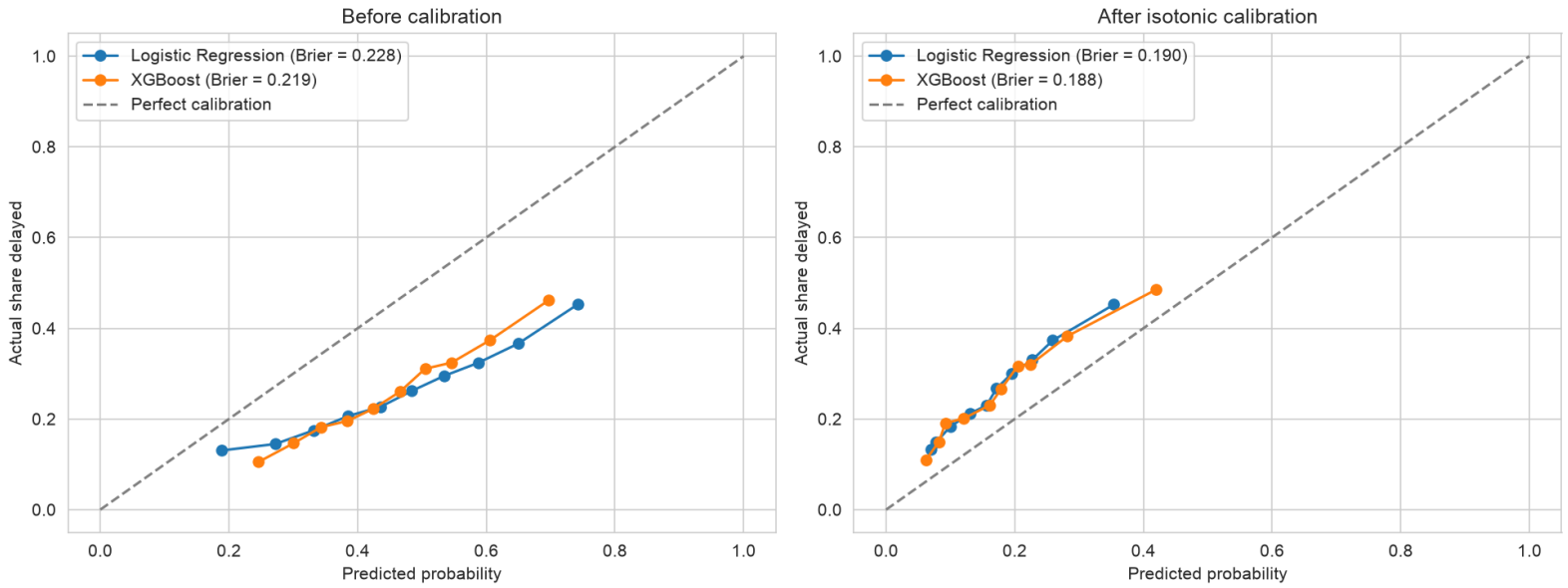
of delayed flights caught when half of the alerts must be real delays (Logistic Regression: 5.9%)

Precision-recall and ROC curves on the test set



Dashed lines show a random guess. Both models are clearly above it, but delays stay hard to predict before departure: weather and air traffic control are not in the features.

Calibration fixes over-confident probabilities



Isotonic regression is fitted on the validation set and applied to the test set.

Brier score (lower is better)

Logistic Regression

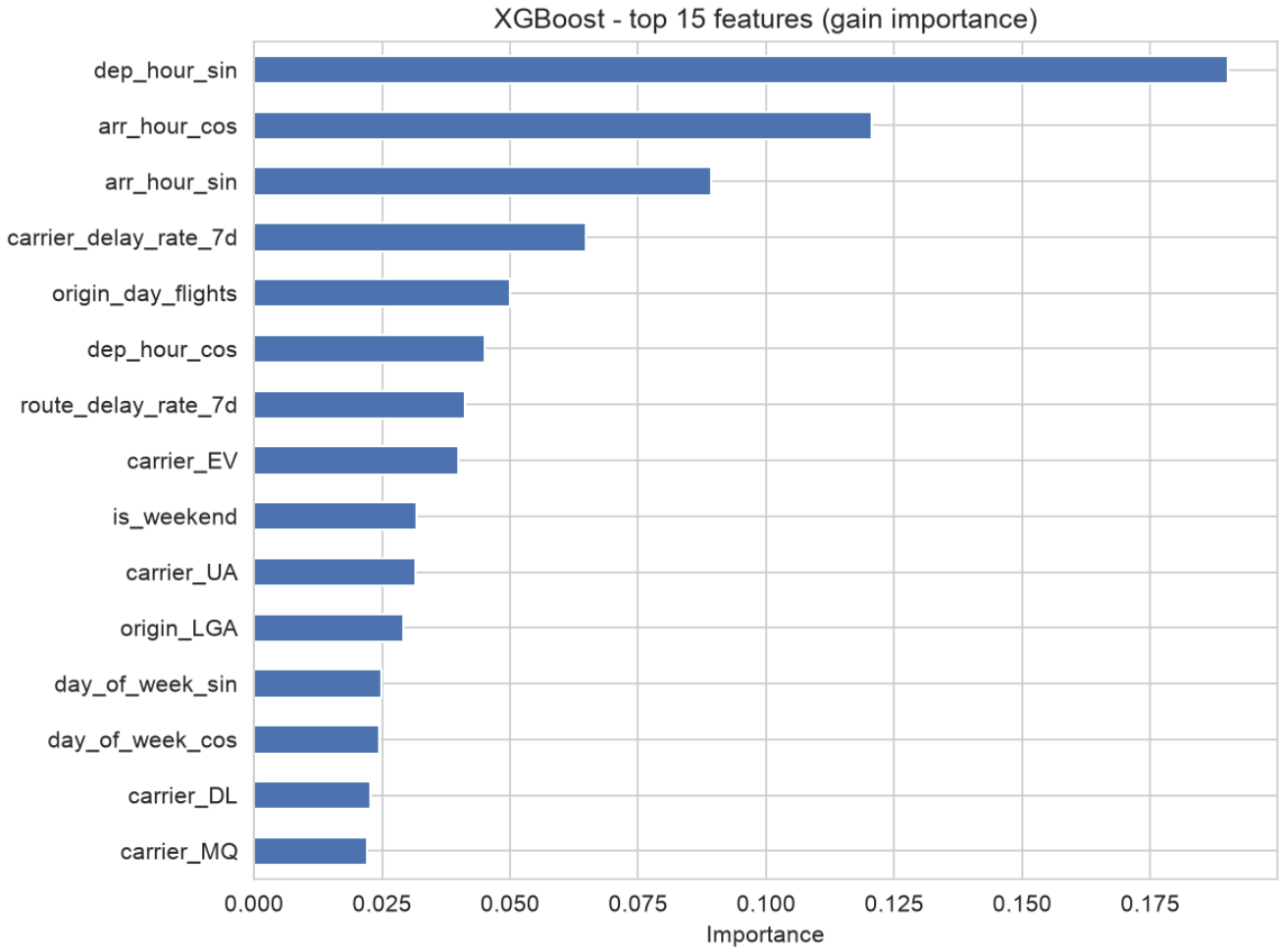
0.228 → 0.190

XGBoost

0.219 → 0.188

A constant guess scores 0.192. The calibrated models slightly under-predict, because Sep to Oct were calmer than Nov to Dec.

Time of day and recent delays drive predictions



Hour of day comes first

The sin and cos of the departure and arrival hour rank highest: later flights are riskier

Recent delays carry over

The carrier's and route's delay rate over the last 7 days

Busy days add risk

More flights scheduled at the origin that day means more delays

Logistic Regression's largest coefficients agree.

Takeaways and next steps

What we found

- XGBoost beats Logistic Regression, most clearly at high precision
- Calibration makes the probabilities usable
- Time of day and recent delays are the strongest signals

Limitations

- Weather and air traffic control are not in the data
- Only NYC airports and only 2013
- Test months differ from training months

Phase 2

- Neural network with learned embeddings for carrier, origin and destination
- Same date split and metrics for a fair comparison

Thank you



Questions?

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github.com/mayank-pillai-99/flight-delay-prediction